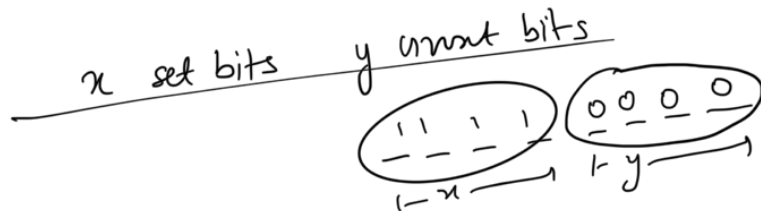


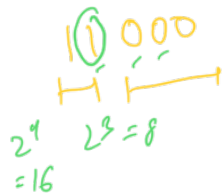
Bit Manipulation 3

Q) Construct a binary number



$x=2, y=3$

$2^3 + 2^4$



2^4 ☆

$x=3, y=2$

$2^2 + 2^3 + 2^4$

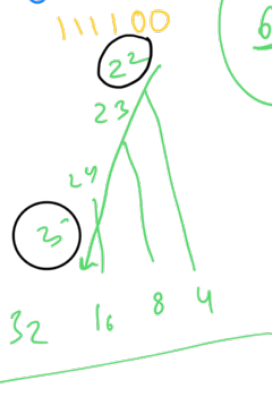


2^8 ☆

$x=4, y=2$

$2^2 + 2^3 + 2^4 + 2^5$

2	(2^2)
3	(2^3)
4	(2^4)
5	(2^5)



60 ☆

$2^2 + 2^3 + 2^4 + 2^5$

ans = 0

$$O(x+y)$$



$$O(x)$$

```

for (i = y; i ≤ y+x-1, i++)
    ans = ans + pow(2, i)
return ans

```

Diagram showing the expansion of the sum: $2^y + 2^{y+1} + \dots + 2^{y+x-1}$

Q.P

$$2^y \quad \dots \quad 2^{y+x-1}$$

$$c \cdot d = r = 2$$

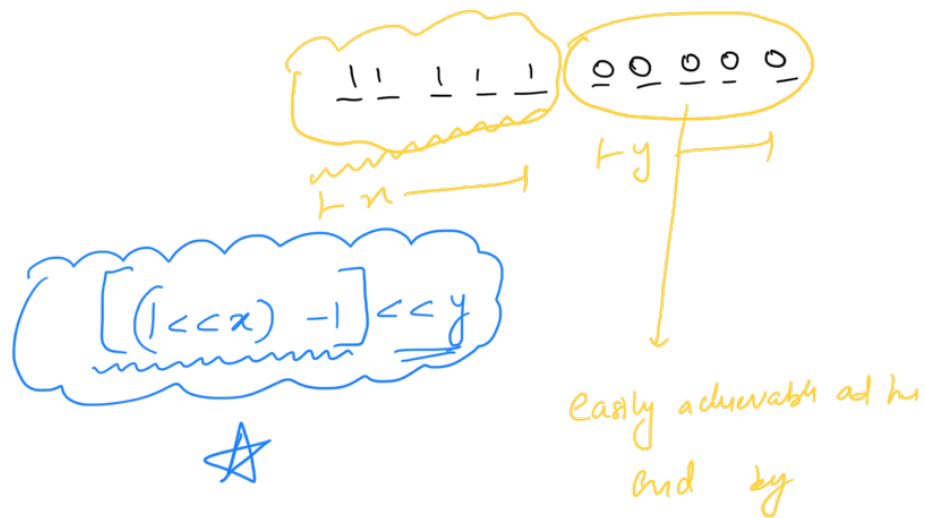
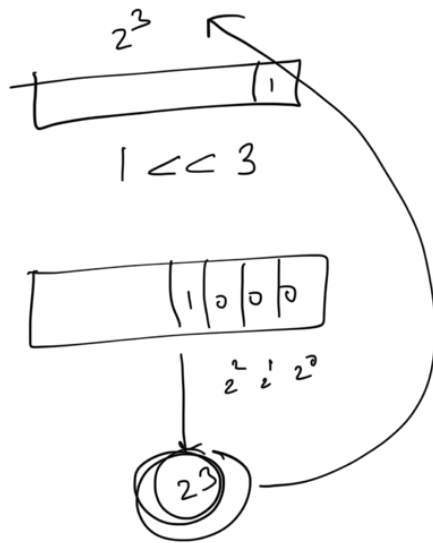
$$S_{G.P} = \frac{a(2^n - 1)}{2 - 1}$$

#km

$$= \frac{2^y (2^x - 1)}{2 - 1}$$

$$= 2^y (2^x - 1)$$

Q1) $\{ (1 < y) \times (1 < x - 1) \}$



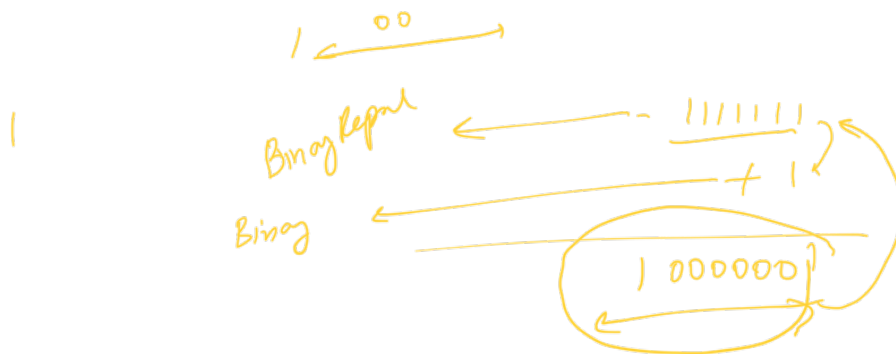
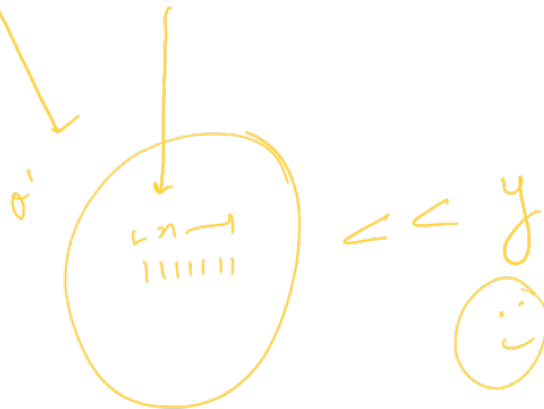
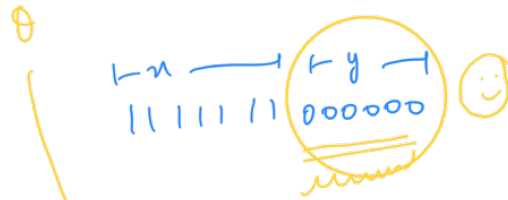
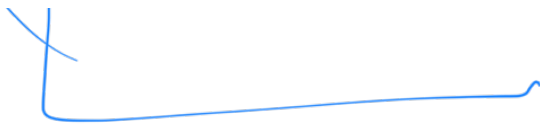
$\ll$
 left shifts y times

$x = 4$
 $x < 6$

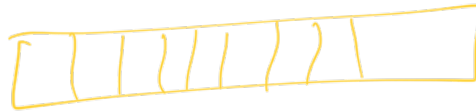
1 1 1 1
 1 1 1 1 1 1

10000
 - 1
 01111





$(1 < 2)$
 print 11111
 print(1 < 2) 4
~~100~~



⑥ indices A, B
 Find a number X
 s.t. $A \wedge X \Rightarrow \text{minimum}$

$A = 3$
 11
 X

you must have ==
B set bits

Given A
 Find X

s.t. $A \wedge X = \text{minimum}$

$A = 10$

1010

$X = \boxed{1010}$

minimal

$A \wedge X \Rightarrow \text{min}$



$A \wedge A = 0$

find $\wedge$



Condition

$X \equiv$ must have
B set bits

A Obs: Use B set bits judiciously

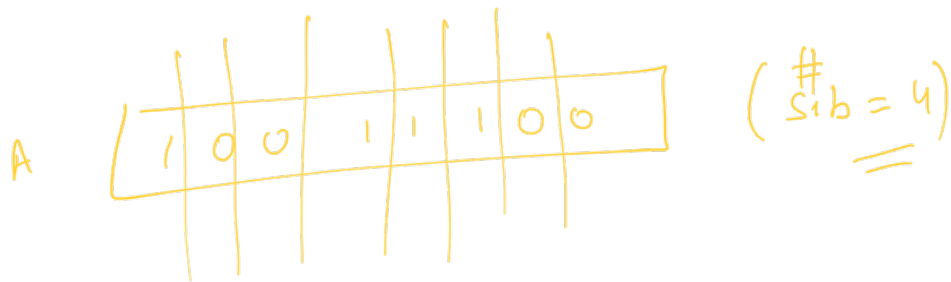
A = 1011000
 (1)

B Obs: Ones must be tried to be used
alongside 1s of A

C Obs: If it is impossible to have
all 1s of X correspond to 1s
of A. Then minimize the damage.

A = 110
B = 5
X

place 1s on the right



X
(5 set bits)

if ($B > \#sb(A)$)
then you need to
increase the delay

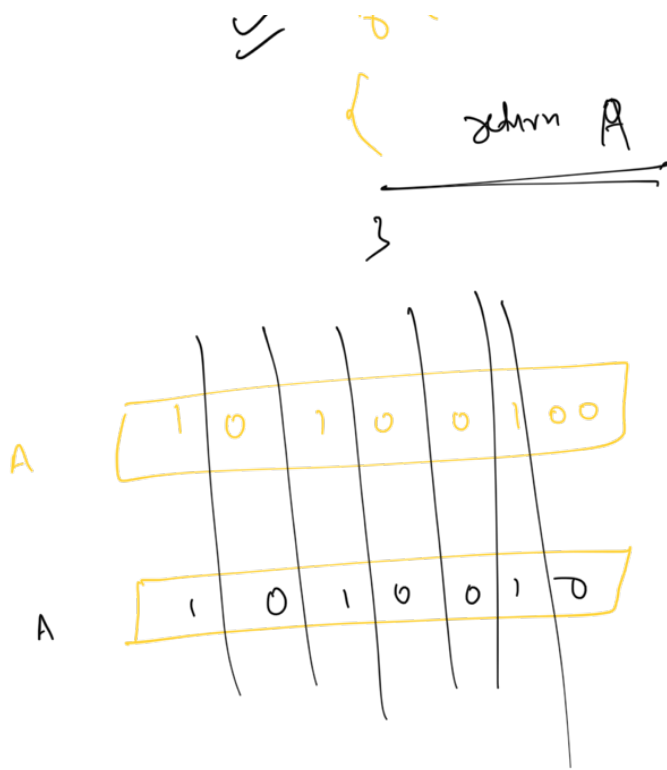
Input, A

--	--	--	--	--	--	--	--

Count $\#sb(A) =$ num of ~~set~~ bits in input
 $=$ set BC in input

B set bits in X, answer

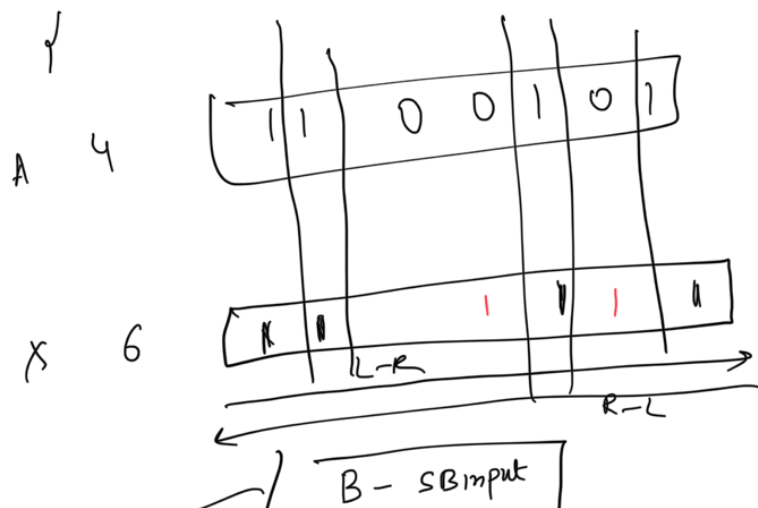
if ($sbc_{input} == B$)



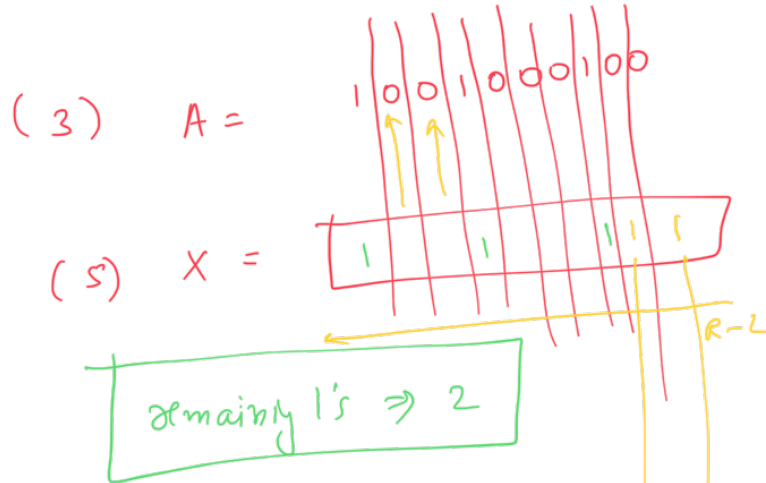
If the number of set bits was same in the
 input and the output

$\Rightarrow \text{output} = \text{input}$

if $(SB_{input} < B)$



Shl need
to array



2^m

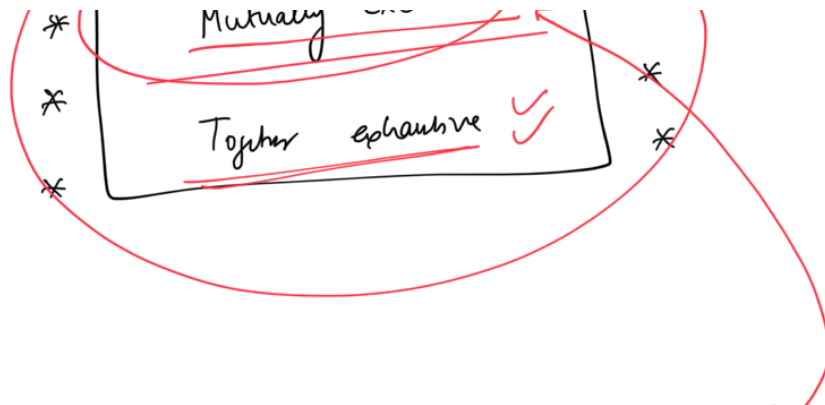


2^{m+1}

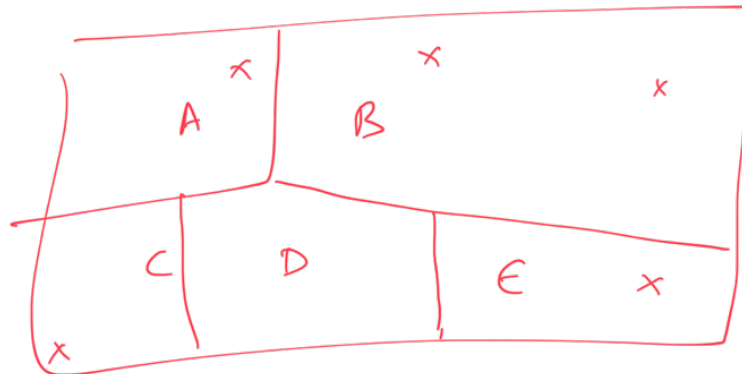
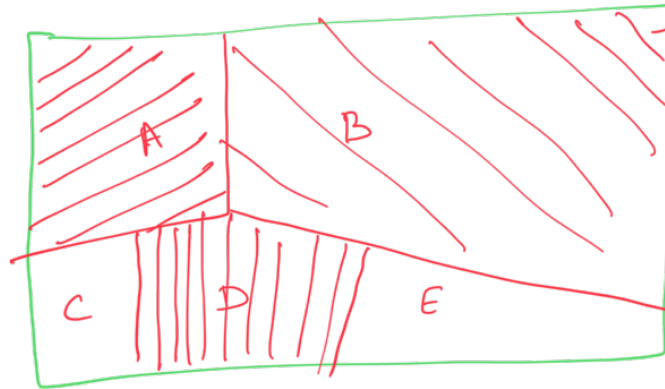


To get the minimal answer
A) enough the pretty 1's from the
right side

... .. exclusive *



possibilities

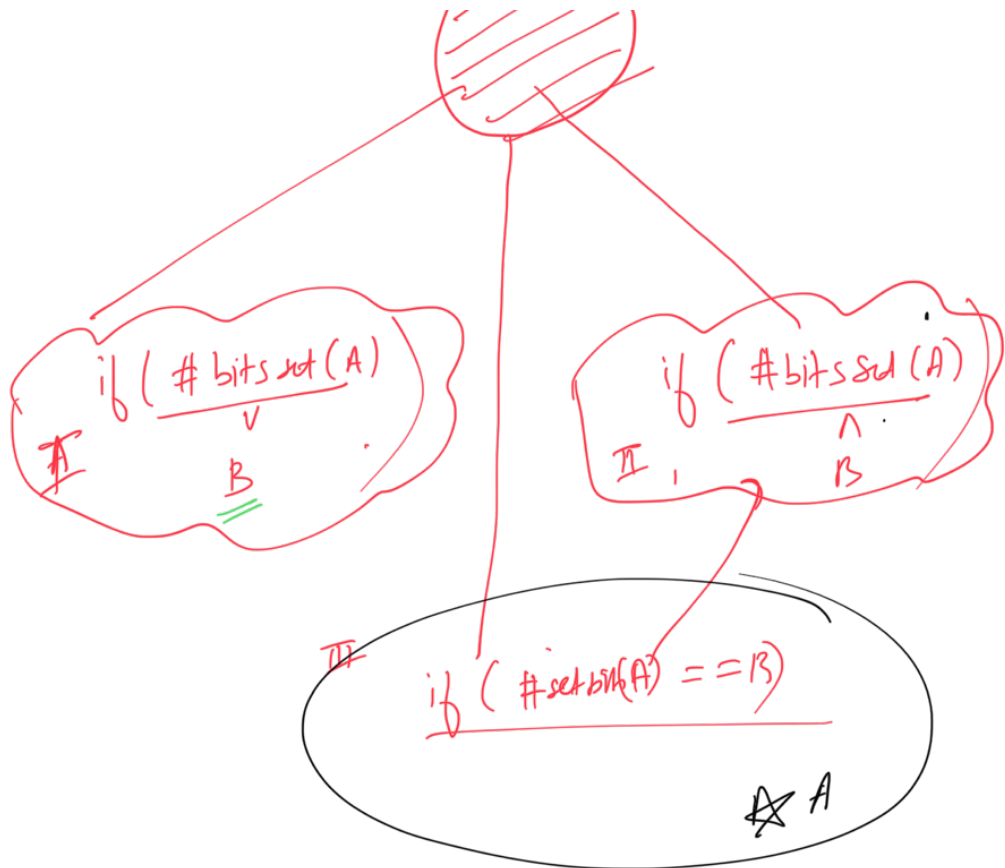


A

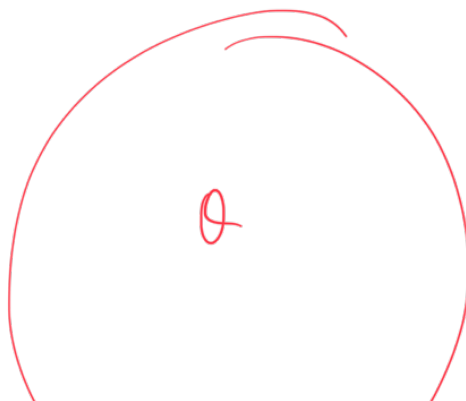
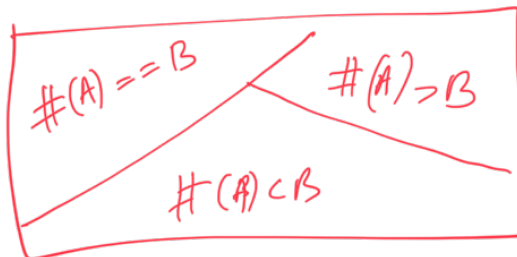
(B sh) x

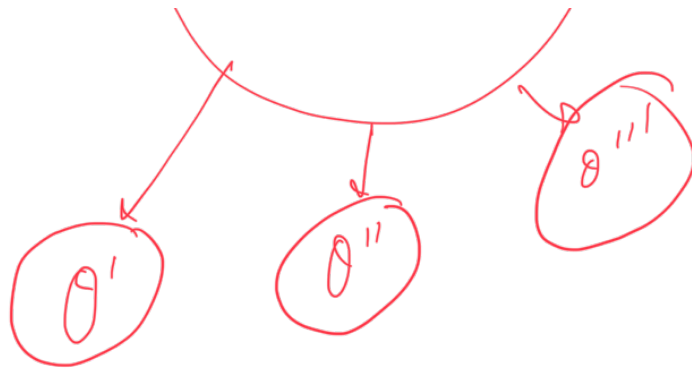
min XOR





together exhaustive





A

1	0	1	1	0	0
---	---	---	---	---	---

(s)B

1		1	1	1	1
---	--	---	---	---	---

LSB part

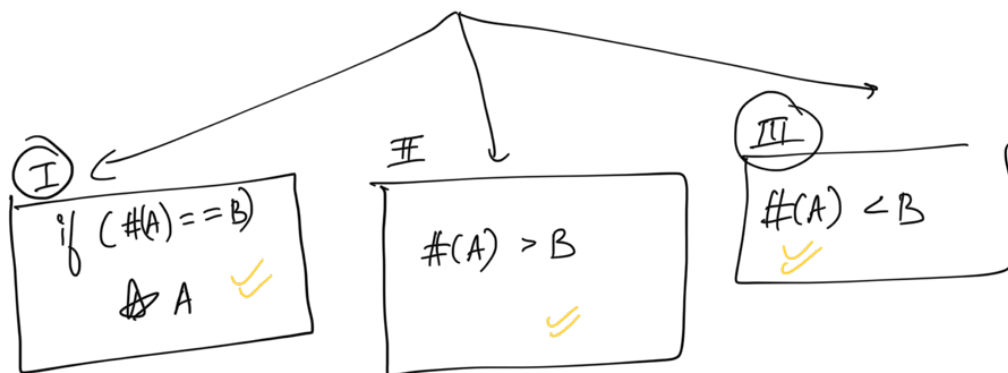
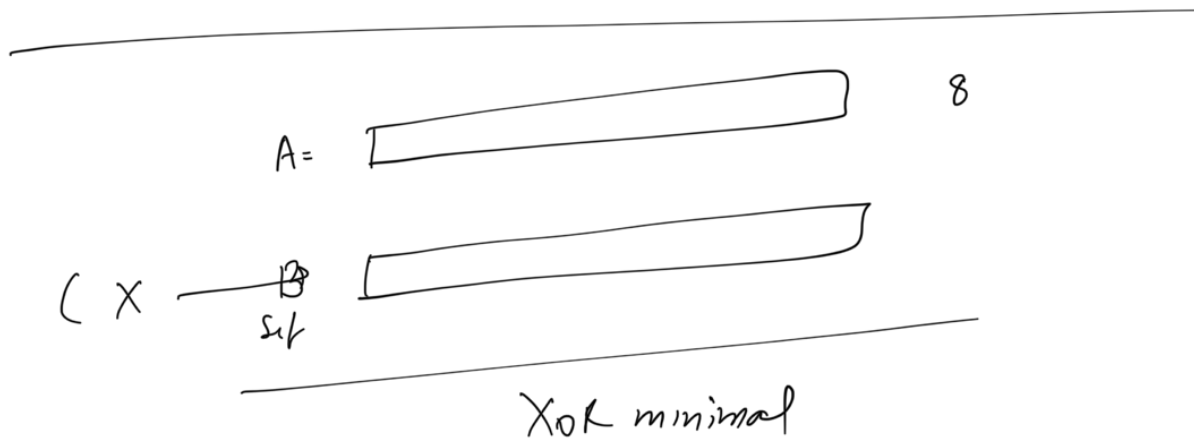
A = (8)

1	0	0	1	1	1	0	0	0	1	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---

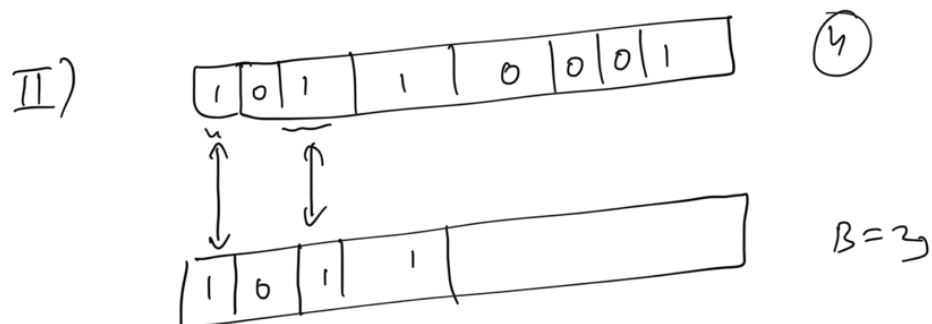
X(s)

1			1	1	1			1				
---	--	--	---	---	---	--	--	---	--	--	--	--

0 0 0 0 0 0 0 0 1 1 0



I) → solved A



XOR minimal

III



(3)



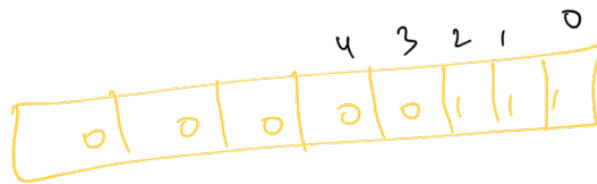
(5)

-3
(2)

0 0 0 0 0 1 1



A



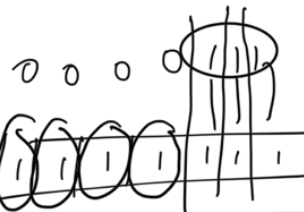
(3)



(5)
-3
(2)



A



(3)

(7)

-3
(4)



Code

①

Count no of set bits in A

②

if else ladder to

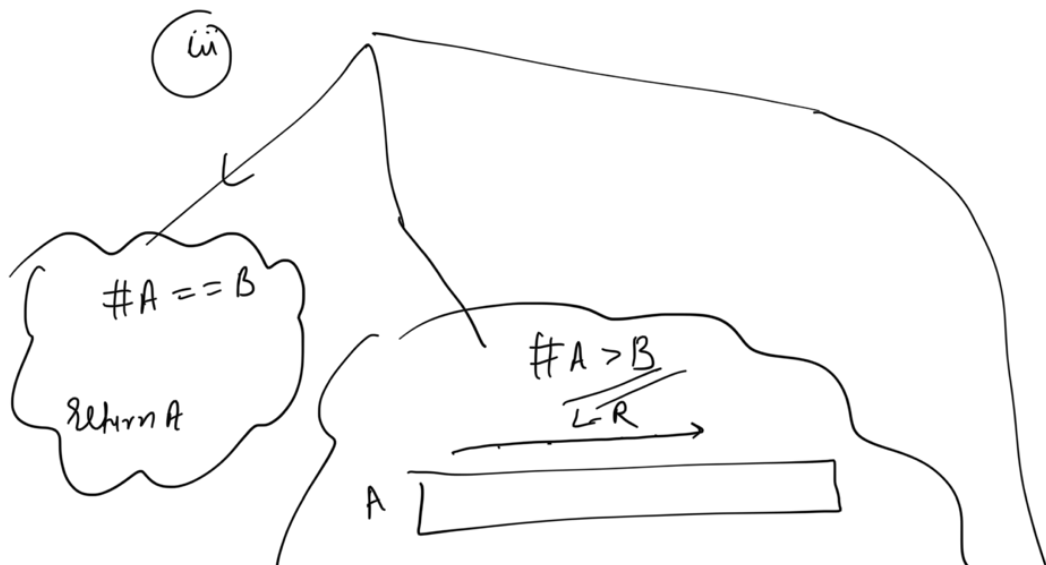
apart

3 subsets of possibilities

if ($\#(A) == B$)

elif ($\#(A) > B$)

else ($\#(A) < B$)

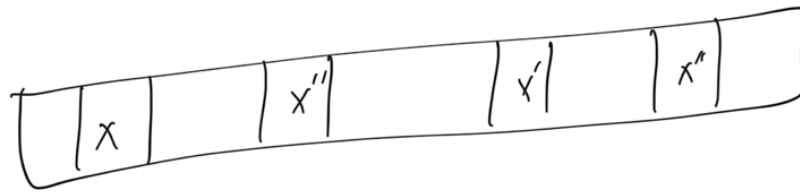
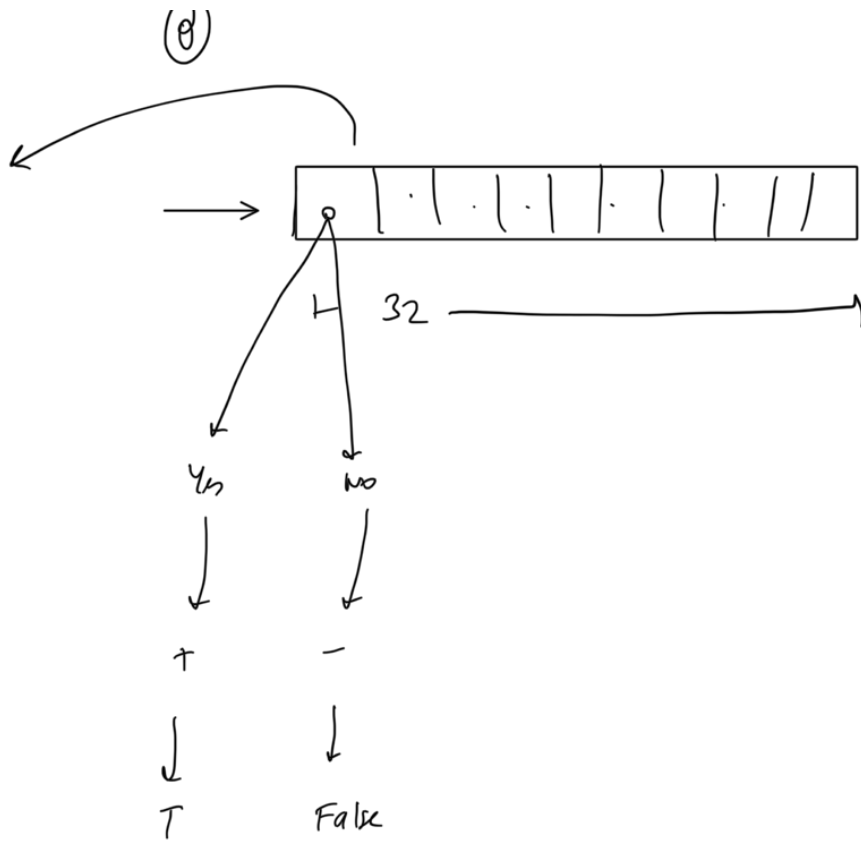


X ing bit shift
 1's at required positions
 Return X

$\#A < B$
 A
 ← set bits
 ↓
 keep moving an
 in X ing bit shift op
 X

For handy loop
 write after loop
 from $R \rightarrow L$
 and set for
 LSR possibly

Return X



W

$$A = \begin{matrix} 111 \\ \cancel{111} \end{matrix} \quad (x)$$

$$B = 1 \underline{1} 111$$

